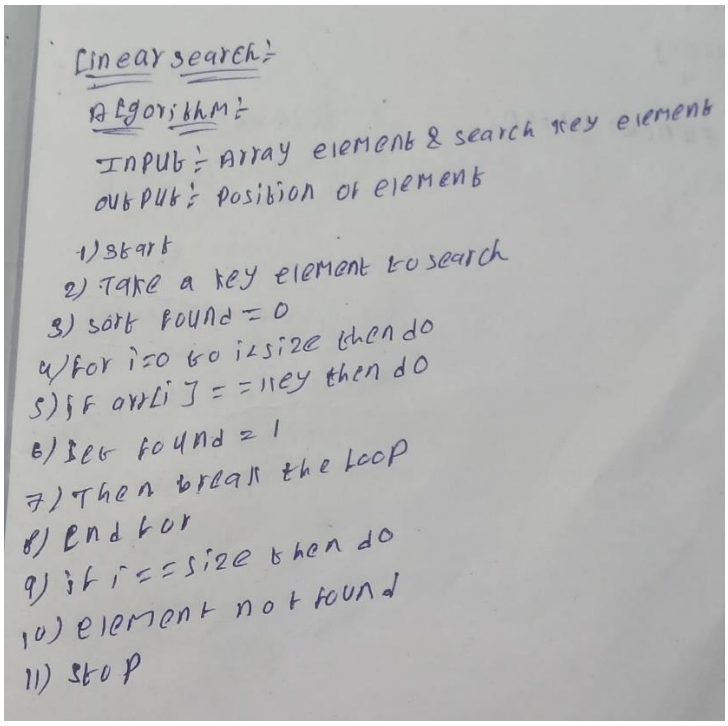
PRACTICAL-02

AIM: Implementation and Time analysis of linear and binary search algorithm.

1.) LINEAR SEARCH

ALGORITHM:



PROGRAM:

```
#include <iostream>

using namespace std;

int main() {

   int arr[20],n,key,i;

   cout << "Enter the number of elements: ";

   cin >> n;

   cout << "Enter the array elements: ";

   for (int i = 0; i < n; i++) {

      cin >> arr[i];

   }
```

```
    cout << "Enter the element to search: ";
    cin >> key;
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            cout << "Element found at index " << i;
            return 0;
        }
    }
    cout << "Element not found";
    return 1;
}
```

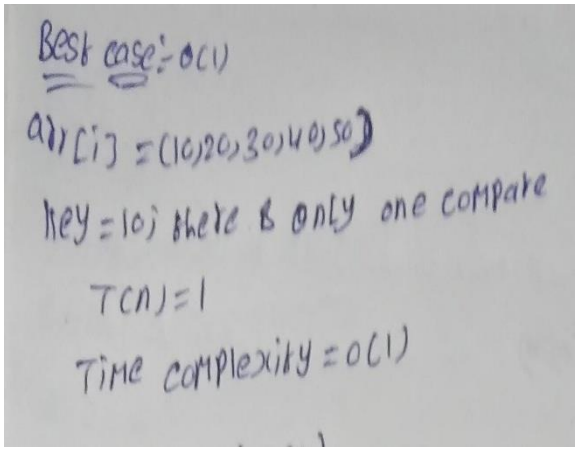
OUTPUT:

```
Output
Enter the number of elements: 5
Enter the array elements: 9
1
6
7
2
Enter the element to search: 2
Element found at index 4

=== Code Execution Successful ===
```

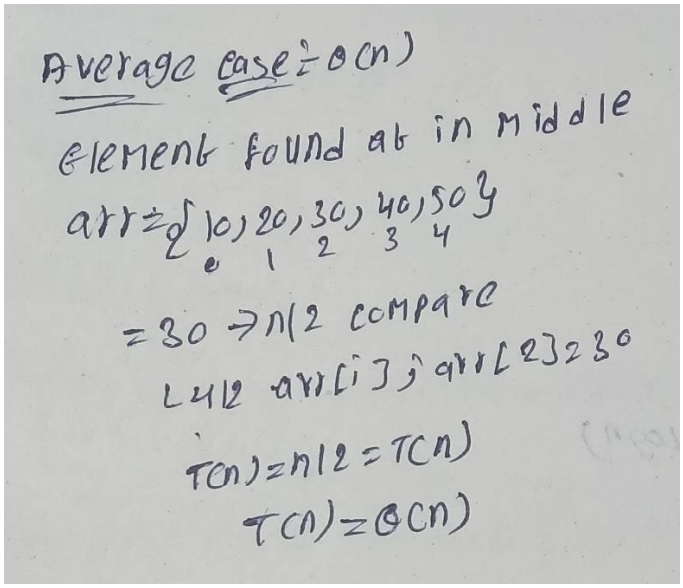
TIME COMPLEXITY:

Best Case:

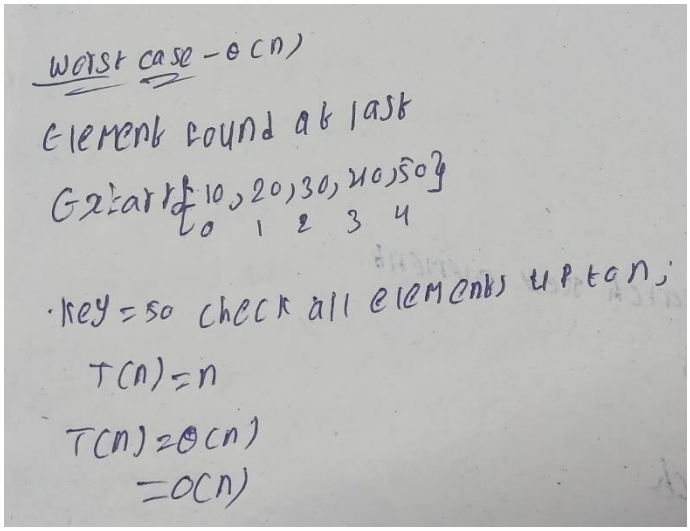


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Average Case:

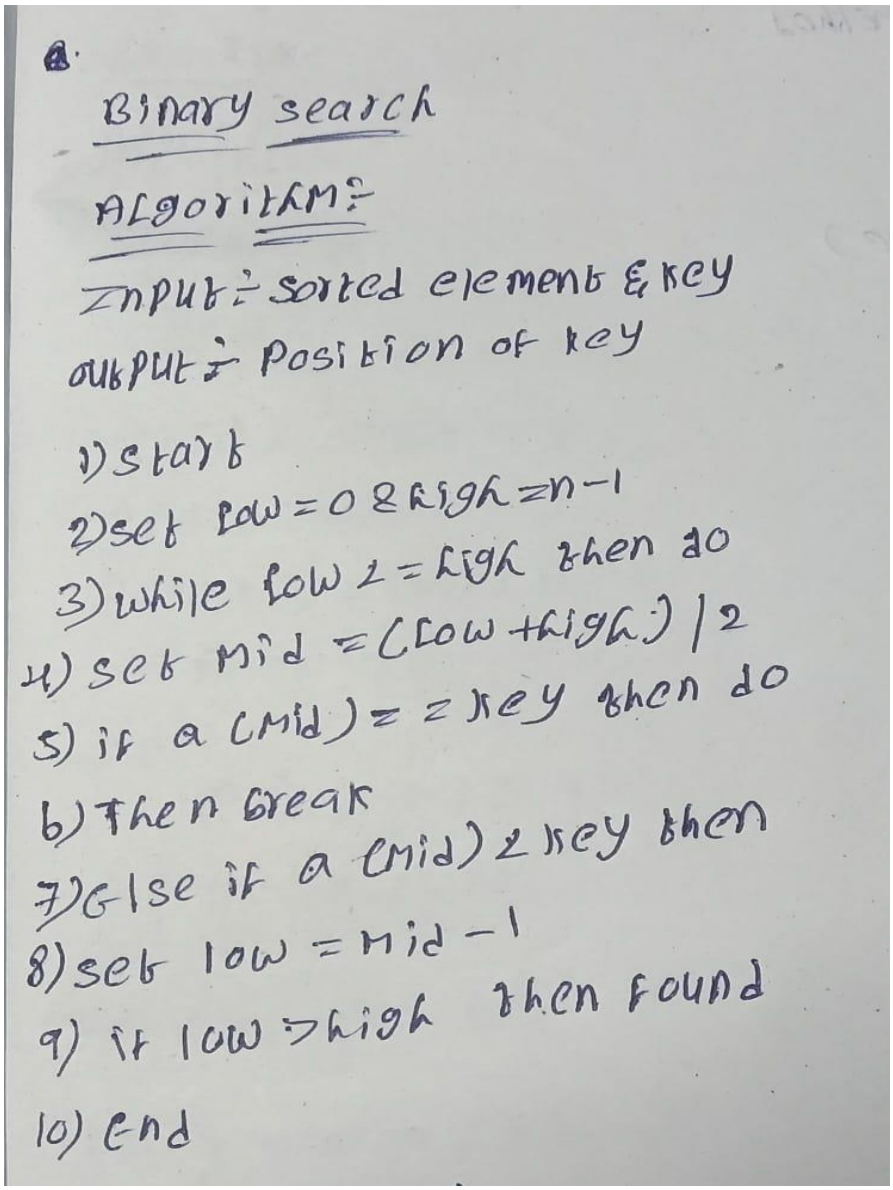


Worst Case:



2.) BINARY SEARCH

ALGORITHM:



PROGRAM:

```
#include <iostream>

using namespace std;

int main()

{

   int n, key;

   int low, high, mid;


   cout << "Enter the number of elements: ";

   cin >> n;
```

```
int a[n];
cout << "Enter the sorted array elements: ";
for (int i = 0; i < n; i++)
{
   cin >> a[i];
}
cout << "Enter the key element: ";
cin >> key;
low = 0;
high = n - 1;
while (low <= high)
{
   mid = (low + high) / 2;
   if (a[mid] == key)
   {
      cout << "Element found at position " << mid << endl;
      break;
   }

   else if (a[mid] < key)
   {
      low = mid + 1;
   }

   else
   {
      high = mid - 1;
   }
}
if (low > high)
{
   cout << "Element not found" << endl;
}
return 0;
```

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}

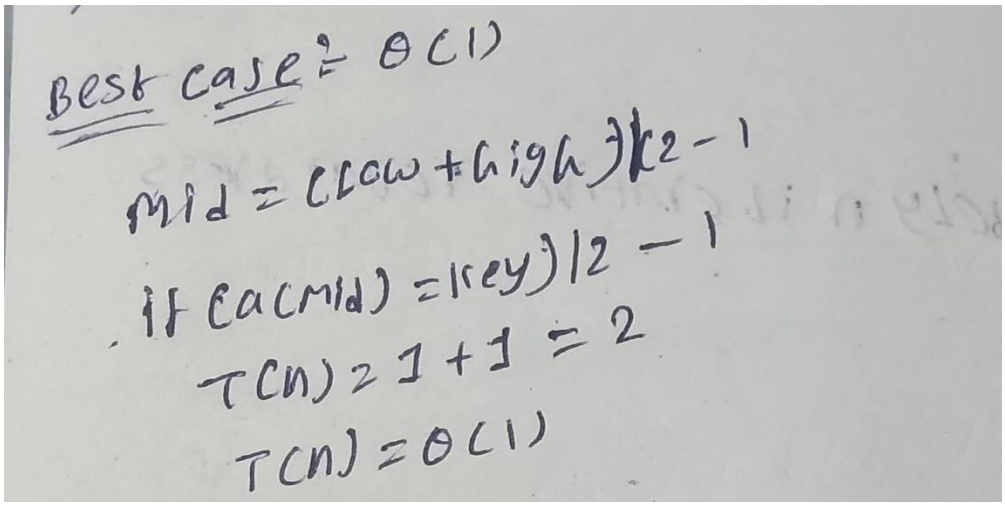
OUTPUT:

```
Output
Enter the number of elements: 5
Enter the sorted array elements: 2
9
15
19
26
Enter the key element: 26
Element found at position 4

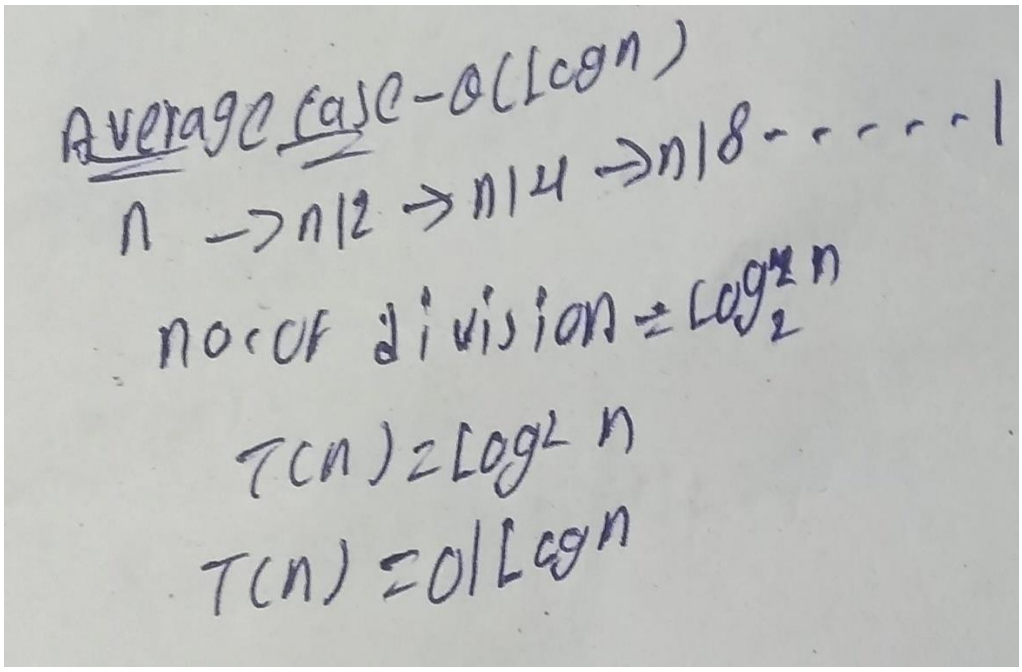

=== Code Execution Successful ===
```

TIME COMPLEXITY:

Best Case:



Average Case:



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Worst Case:

